

# Step 1 - Understand the Problem and Establish Design Scope

Designing a cloud storage system like [Google Drive](#) is a large-scale system design problem. Therefore, the first step is to clarify requirements and narrow the scope of the discussion.

Below is an example of a candidate-interviewer conversation:

**Candidate:** What are the most important features?

**Interviewer:** Upload and download files, file sync, and notifications.

**Candidate:** Is this a mobile app, a web app, or both?

**Interviewer:** Both.

**Candidate:** What are the supported file formats?

**Interviewer:** Any file type.

**Candidate:** Do files need to be encrypted?

**Interviewer:** Yes, files in storage must be encrypted.

**Candidate:** Is there a file size limit?

**Interviewer:** Yes, files must be 10 GB or smaller.

**Candidate:** How many users does the product have?

**Interviewer:** 10 million DAU.

## Features in Scope

In this chapter, we focus on the following core features:

### 1. File Upload

Users can upload files into the cloud storage system.

The easiest and most common method is drag-and-drop upload from desktop or mobile devices.

### 2. File Download

Users should be able to download stored files from anywhere and on any supported device.

### **3. File Synchronization**

When a file is added or modified on one device, it should automatically synchronize across all other connected devices.

### **4. File Revision History**

Users should be able to see older versions of files and restore previous revisions if needed.

### **5. File Sharing**

Users can share files with friends, family members, or coworkers through permissions or shared links.

### **6. Notifications**

The system should notify users when:

- A file is edited
- A file is deleted
- A file is shared with them

## **Features Out of Scope**

The following advanced features are not covered in this design:

### **Real-Time Document Collaboration**

Applications like Google Docs allow multiple users to edit the same document simultaneously.

This requires technologies such as:

- Operational Transformation (OT)
- Conflict-free Replicated Data Types (CRDTs)
- Real-time collaboration engines

These are outside the scope of this chapter.

## **Non-Functional Requirements**

Besides functional requirements, understanding non-functional requirements is extremely important.

## **Reliability**

Reliability is critical for any cloud storage system.

- Data loss is unacceptable.
- Files must be replicated and backed up properly.
- The system should support disaster recovery.

## **Fast Synchronization Speed**

File synchronization should happen quickly.

Slow synchronization leads to poor user experience and user dissatisfaction.

## **Efficient Bandwidth Usage**

The system should minimize unnecessary network usage, especially for users on limited mobile data plans.

Techniques such as:

- Delta sync
- Compression
- Chunked upload
- Deduplication

can help reduce bandwidth consumption.

## **Scalability**

The platform must handle:

- Millions of users
- Billions of files
- High concurrent uploads/downloads

The architecture should scale horizontally.

## **High Availability**

Users should still be able to access files even if:

- Some servers fail
- Network issues occur
- Certain regions go offline

This requires:

- Replication
- Failover mechanisms
- Distributed infrastructure

## Final Design Scope Summary

The system we design will support:

- File upload and download
- Cross-device synchronization
- File versioning
- File sharing
- Notifications
- Encryption
- High scalability and availability